

# Geophysical Inversion

## Problem Set Week 1

Magnús Pálsson

### Problem 1.1

Suppose that you determine the masses of 100 objects by weighing the first, then weighing the first and second together, and then weighing the rest in triplets: the first, second, and third; the second, third, and fourth; and so forth.

#### A

Identify the data and model parameters in this problem. How many of each are there?

#### Solution

The data is the measurements of the triples. The model parameters are estimates of the real weight of the 100 objects. There are 100 of each.

#### B

Write down the matrix  $\mathbf{G}$  in the equation  $\mathbf{d} = \mathbf{G}\mathbf{m}$  that relates the data to the model parameters.

#### Solution

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & \dots & \dots & \dots & 0 \\ 1 & 1 & 0 & 0 & & & & & \vdots \\ 1 & 1 & 1 & 0 & & & & & \vdots \\ 0 & 1 & 1 & 1 & & & & & \vdots \\ \vdots & & & & \ddots & & & & \vdots \\ \vdots & & & & & & 1 & 1 & 1 & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 & 1 & 1 & 1 \end{bmatrix}$$

#### C

How sparse is  $\mathbf{G}$ ? What percent of it is zero

### Solution

Approximately 97% is zero. For a more precise estimate that takes the first two lines into account; we have:

$$1 - \frac{\frac{1}{100} + \frac{2}{100} + 98 \times \frac{3}{100}}{100} = 1 - \frac{297}{100^2} = \frac{9703}{10000}$$

This means that 97.03% of the matrix is zero.

## Problem 1.2

Suppose that you determine the height of 50 objects by measuring the first, and then stacking the second on top of the first and measuring their combined height, stacking the third on top of the first two and measuring their combined height, and so forth.

### A

Identify the data and model parameters in this problem. How many of each are there?

### Solution

The data are the height measurements of the different stacks. The model parameters are estimates of the heights of individual objects. There are 50 of each.

### B

Write down the matrix  $\mathbf{G}$  in the equation  $\mathbf{d} = \mathbf{G}\mathbf{m}$  that relates the data to the model parameters.

### Solution

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & 0 & & \vdots \\ 1 & 1 & 1 & 0 & & \vdots \\ 1 & 1 & 1 & 1 & & \vdots \\ \vdots & & & & \ddots & \vdots \\ 1 & 1 & 1 & 1 & \cdots & 1 \end{bmatrix}$$

### C

How sparse is  $\mathbf{G}$ ? What percent of it is zero

### Solution

We can immediately see that slightly less than half of the matrix will be zero. For a more precise estimate we can see that line  $n$  contains the number 1  $n$  times. To get the total number of 1s we can use the formula  $\frac{n(n+1)}{2}$  to add up the numbers from 1 to 50 and then divide by the total number of elements:  $50^2$ .

$$\frac{\frac{50 \times 51}{2}}{50^2} = \frac{1275}{2500} = 0.51$$

Therefore  $100 - 51 = 49\%$  of the matrix is zero.

## Problem 1.3

Write a program to compute  $\mathbf{G}$  in the case of the cubic equation,  $T = a + bz + cz^2 + dz^3$ . Assume that 11  $z$ s are equally spaced from 0 to 10.

### Solution

```
# Compute the G matrix for a cubic polynomial model given vector z.
# z is a vector represented either as a
# 1D array or a 2D array column vector (z.shape[1] == 1)
def ComputeG(z):
    n = z.shape[0]

    #In case z is a 1D array
    if len(z.shape) == 1:
        z = z.reshape(n,1)

    G = np.concatenate((np.ones((n,1)), z, z**2, z**3), axis = 1)
    return G

z = np.arange(11)
print(ComputeG(z))
```

Output:

```
[[ 1.  0.  0.  0.]
 [ 1.  1.  1.  1.]
 [ 1.  2.  4.  8.]
 [ 1.  3.  9. 27.]
 [ 1.  4. 16. 64.]
 [ 1.  5. 25. 125.]
 [ 1.  6. 36. 216.]
 [ 1.  7. 49. 343.]
 [ 1.  8. 64. 512.]
 [ 1.  9. 81. 729.]
 [ 1. 10. 100. 1000.]]
```

## Problem 1.4

Let the data  $\mathbf{d}$  be the running average of the model parameters,  $\mathbf{m}$ , computed by averaging groups of three neighboring points; that is,  $d_i = (m_{i-1} + m_i + m_{i+1})/3$

### A

What is the matrix  $\mathbf{G}$  in the equation  $\mathbf{d} = \mathbf{Gm}$  in this case?

### Solution

The matrix  $\mathbf{G}$  is the matrix with zeros almost everywhere except around the diagonal where the number  $\frac{1}{3}$  appears three times (See section B for details).

### B

What problems arise at the top and bottom rows of the matrix and how can you deal with them?

### Solution

The definition of the matrix runs into problems at the boundaries where either index,  $i - 1$  or  $i + 1$  are out of bounds and the corresponding parameter doesn't exist. There are two straightforward solutions.

#### Possibility 1

We take the average of the available parameters, giving us:

$$\begin{aligned}d_1 &= (m_1 + m_2)/2 \\ d_N &= (m_{N-1} + m_N)/2\end{aligned}$$

Where  $N$  is the total number of parameters, which will be equal to the total number of datapoints for this example. This yields the matrix:

$$\mathbf{G} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & \dots & \dots & \dots & \dots & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & & & & & \vdots \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & & & & & \vdots \\ \vdots & & & & \ddots & & & & \vdots \\ \vdots & & & & & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \vdots & & & & & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & \dots & \dots & \dots & \dots & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

#### Possibility 2

We simply ignore the boundary. This is for example often done in image processing when applying a convolutional filter. This means that  $\mathbf{d}$  will have 2 fewer elements than  $\mathbf{m}$  which might not be desirable. This approach would yield the matrix

$$\mathbf{G} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & & & & & \vdots \\ \vdots & & & & \ddots & & & & \vdots \\ \vdots & & & & & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

### C

How sparse is  $\mathbf{G}$ ? What percent of it is zero?

### Solution

The precise percentage depends on the solution to the boundary problem but it will be around  $\frac{N-3}{N} \times 100\%$

## Problem 1.5

Simplify Eq. (1.20) by assuming that there is only one sample  $S_{1j}$  whose composition is measured. Consider the case where the composition of the  $p$  factors is known, but their proportions in the sample are unknown, and rewrite Eq. (10.2) in the form  $\mathbf{d} = \mathbf{G}\mathbf{m}$ .

## Solution

Looking at the matrix form of Eq. (1.20)  $\mathbf{S} = \mathbf{C}\mathbf{F}$

$$\begin{bmatrix} S_{11} & & & \\ & \ddots & & \\ & & S_{ij} & \\ & & & \ddots \\ & & & & S_{NM} \end{bmatrix} = \begin{bmatrix} C_{11} & & & \\ & \ddots & & \\ & & C_{ik} & \\ & & & \ddots \\ & & & & C_{Np} \end{bmatrix} \begin{bmatrix} F_{11} & & & \\ & \ddots & & \\ & & F_{kj} & \\ & & & \ddots \\ & & & & F_{pM} \end{bmatrix}$$

If we have  $N = 1$  then this turns into

$$\begin{bmatrix} S_{11} & \cdots & S_{1M} \end{bmatrix} = \begin{bmatrix} C_{11} & \cdots & C_{1k} & \cdots & C_{1p} \end{bmatrix} \begin{bmatrix} F_{11} & & & \\ & \ddots & & \\ & & F_{kj} & \\ & & & \ddots \\ & & & & F_{pM} \end{bmatrix}$$

Using the identity relating taking the transpose of a matrix and matrix multiplication;  $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ , we get

$$\begin{bmatrix} S_{11} \\ \vdots \\ S_{j1} \\ \vdots \\ S_{M1} \end{bmatrix} = \begin{bmatrix} F_{11} & & & \\ & \ddots & & \\ & & F_{jk} & \\ & & & \ddots \\ & & & & F_{Mp} \end{bmatrix} \begin{bmatrix} C_{11} \\ \vdots \\ C_{k1} \\ \vdots \\ C_{p1} \end{bmatrix}$$

This gives us the form we are looking for with  $\mathbf{d} = \mathbf{S}$ ,  $\mathbf{G} = \mathbf{F}$ ,  $\mathbf{m} = \mathbf{C}$